运行环境

组件	版本
Java	**JDK 8 (1.8)**
Hadoop	**3.3.6**
Hive	**3.1.3**
MySQL	**8.0**
Python	**3.10（系统自带）**

环境目录

基础环境

```
/opt/env
├── java/                # JDK安装目录
├── python/              # Python环境
│   └── venvs/           # 虚拟环境
│       └── marketing_dw
└── mysql/               # MySQL（客户端工具/说明）
```

虚拟环境位置

```
/opt/env/python/venvs
```

大数据组件

```
/opt/bigdata
├── hadoop/            # Hadoop安装目录
├── hive/              # Hive安装目录
├── spark/             # Spark安装目录（可选）
└── zookeeper/         # 预留（可不装）
```

安装包归档

```
/opt/pkg
├── jdk/
├── python/
├── hadoop/
├── hive/
├── spark/
└── mysql/
```

软件“使用目录”

Hadoop 使用目录

```
/data/hadoop
├── hdfs
│   ├── namenode/      # NameNode数据
│   └── datanode/      # DataNode数据
├── tmp/               # 临时目录
└── logs/              # Hadoop日志
```

Hive 使用目录

```
/data/hive
├── warehouse/         # Hive仓库（最重要）
├── tmp/               # 临时数据
└── logs/              # Hive日志
```

Spark 使用目录

```
/data/spark
├── tmp/
└── logs/
```

MySQL 数据目录

```
/data/mysql
```

环境变量

```
# 基础环境
export JAVA_HOME=/opt/env/java

# Hadoop
export HADOOP_HOME=/opt/bigdata/hadoop

# Hive
export HIVE_HOME=/opt/bigdata/hive
# 项目路径
export PROJECT_HOME=/workspace/marketing_dw
export DATA_HOME=/data

# 正确写法：保留系统 PATH，再追加自己的
export PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/b
in:/sbin:/bin:$JAVA_HOME/bin:$HADOOP_HOME/bin:$HADOOP_HOME/
sbin:$HIVE_HOME/bin

（可选）
# Python虚拟环境
source /opt/env/python/venvs/marketing_dw/bin/activate
```

```
source /workspace/marketing_dw/conf/env.sh
```

hadoop-env.sh配置

```
export JAVA_HOME=/opt/env/java

export HADOOP_MAPRED_HOME=/opt/bigdata/hadoop
export HADOOP_COMMON_HOME=/opt/bigdata/hadoop
export HADOOP_HDFS_HOME=/opt/bigdata/hadoop
export YARN_HOME=/opt/bigdata/hadoop
```

hdfs-site.xml配置

```
vi /opt/bigdata/hadoop/etc/hadoop/hdfs-site.xml
```

```
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>

  <property>
    <name>dfs.namenode.name.dir</name>
    <value>file:/data/hadoop/hdfs/namenode</value>
  </property>

  <property>
    <name>dfs.datanode.data.dir</name>
    <value>file:/data/hadoop/hdfs/datanode</value>
  </property>
</configuration>
```

core-site.xml配置

```
vi /opt/bigdata/hadoop/etc/hadoop/core-site.xml
```

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>

  <property>
    <name>hadoop.tmp.dir</name>
    <value>/data/hadoop/tmp</value>
  </property>
</configuration>
```

hive-site.xml配置

```
<configuration>
  <property>
    <name>javax.jdo.option.ConnectionURL</name>
    <value>jdbc:mysql://localhost:3306/hive_metastore?creat
eDatabaseIfNotExist=true</value>
  </property>

  <property>
    <name>javax.jdo.option.ConnectionDriverName</name>
    <value>com.mysql.cj.jdbc.Driver</value>
  </property>

  <property>
    <name>javax.jdo.option.ConnectionUserName</name>
    <value>hive</value>
  </property>

  <property>
    <name>javax.jdo.option.ConnectionPassword</name>
    <value>hive123</value>
  </property>

  <property>
```

```
    <name>hive.metastore.warehouse.dir</name>
    <value>/data/hive/warehouse</value>
  </property>
</configuration>
```

mapred-site.xml

```
<configuration>
  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>
</configuration>
```

yarn-env.sh

```
export HADOOP_MAPRED_HOME=/opt/bigdata/hadoop
```

yarn-site.xml

```
<configuration>
  <property>
    <name>yarn.resourcemanager.hostname</name>
    <value>localhost</value>
  </property>

  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>
</configuration>
```